

PROJECT PROPOSAL

Title

Development and Implementation of a Digital Registry Management, Office Directory and Visitor Guidance System for the Tanzania Revenue Authority (TRA)

1.0 BACKGROUND OF THE PROJECT

The Tanzania Revenue Authority (TRA) is responsible for administering and collecting government revenue through the implementation of various tax and customs laws and the provision of services to taxpayers and other stakeholders. In carrying out its daily activities, TRA offices handle a large amount of information, correspondence, files, applications, reports and other official documents. The Registry/Records Office plays an important role in receiving, registering, distributing, tracking, storing and retrieving these documents.

Traditionally, many registry activities have depended heavily on manual procedures, including physical files, paper registers, shelves, movement registers and face-to-face inquiries. Although these methods can support office operations, they can also create difficulties when the volume of records becomes large.

Another challenge is that visitors and taxpayers may not always know which office provides a particular service, the office number, floor, location, required procedure or documents needed. As a result, visitors may have to repeatedly ask Registry or reception staff for directions.

This proposal therefore recommends the development of an integrated Digital Registry Management, Office Directory and Visitor Guidance System.

The proposed system will combine digital registry operations with a large information screen/kiosk that displays registry activities, office locations, available services and procedures. It can also provide QR codes through which visitors can access the same information using their mobile phones.

2.0 PROBLEM STATEMENT

The current manual approach to registry and visitor guidance can create several operational challenges.

These include:

- *Large volumes of physical documents.
- *Difficulty locating files quickly.
- *Delays in distributing correspondence.

- *Difficulty tracking the movement of physical files.
- *Risk of misplaced or damaged documents.
- *High dependence on paper records.
- *Limited real-time visibility of document status.
- *Difficulty maintaining updated office directories.
- *Visitors having insufficient information about where services are provided.
- *Visitors repeatedly asking Registry/reception staff for directions.
- *Difficulty knowing the correct procedure for accessing services.
- *Limited ability to generate management reports from manual registers.
- *Risk of unauthorized access to sensitive records.
- *Increased storage requirements for physical files.

Therefore, there is a need for a centralized digital solution that improves document management, office navigation, information access, accountability and service efficiency.

3.0 PROJECT AIM

The main aim of this project is:

To design and implement an integrated digital system that modernizes Registry operations and provides visitors with accurate information about TRA offices, services, locations and procedures.

4.0 SPECIFIC OBJECTIVES

The project will aim to:

- Digitize the registration of incoming and outgoing correspondence.
- Provide electronic document storage and retrieval.
- Track the movement and status of files/documents.
- Reduce dependence on manual registers.
- Improve security and accountability of records.
- Provide a centralized directory of TRA offices.
- Display office numbers, floors and locations.

- Provide information about services offered by each office.
- Provide step-by-step procedures for accessing services.
- Provide an interactive search facility.
- Provide a large digital information screen at Registry/reception.
- Provide QR codes for accessing office information using smartphones.
- Provide management reports and statistics.
- Reduce unnecessary movement and inquiries within the building.

5.0 PROPOSED SOLUTION

The proposed solution is a Digital Registry Management and Office Information System.

The system will consist of two major components:

A. Internal Digital Registry System

Used by authorized TRA staff to manage:

- Incoming correspondence
- Outgoing correspondence
- Digital files
- Document registration
- File movement
- Document tracking
- Scanning
- Classification
- Archiving
- Reports

B. Public Office Directory and Information Display

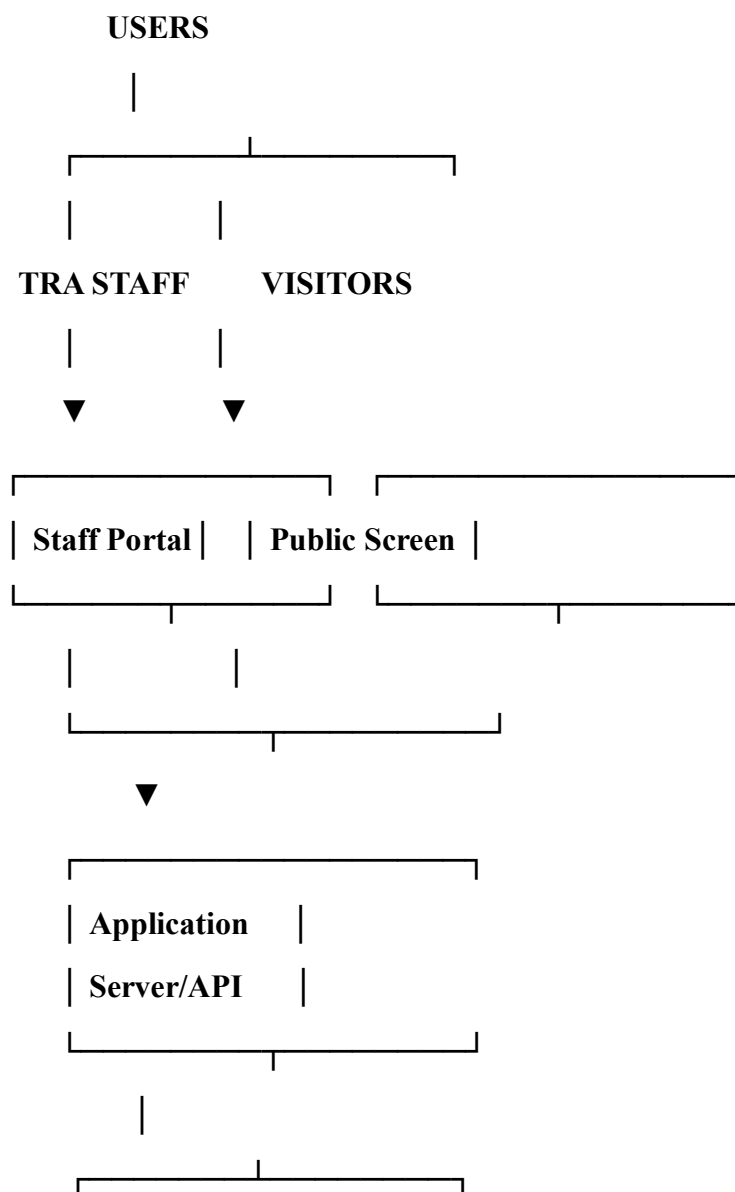
Used by visitors and taxpayers to:

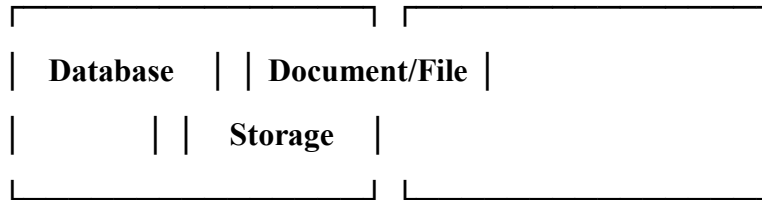
- Search for offices.
- Search for services.
- Find office numbers.

- Find floors and locations.
- View procedures.
- View required documents.
- Follow building directions.
- Scan QR codes.
- View announcements.

6.0 PROPOSED SYSTEM ARCHITECTURE

The system can use a three-tier architecture.





Main components

1. User Interface Layer

- Staff dashboard
- Administrator dashboard
- Public screen
- Search interface

Mobile/QR interface

2. Application Layer

- Authentication
- Registry workflow
- Office directory
- Search engine
- Notifications
- Reporting
- Audit trail

3. Data Layer

- User database
- Document database
- Office database
- Service database
- File movement records
- Audit logs

7.0 SYSTEM WORKFLOW

7.1 Incoming Document Workflow

Document Received



Document Verification



Registration



Reference Number Generated



Scanning/Uploading



Classification



Assign to Department/Officer



Officer Receives



Action Taken



Status Updated



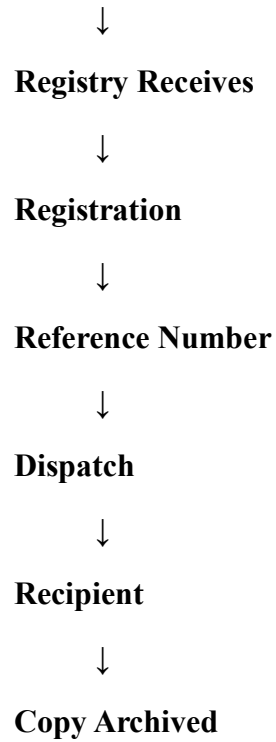
Digital Filing/Archive

7.2 Outgoing Document Workflow

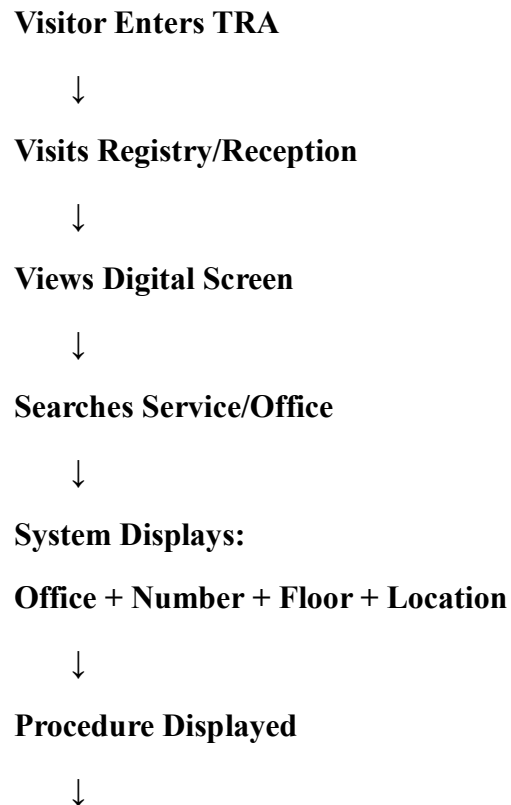
Department Creates Document



Approval



7.3 Visitor Guidance Workflow



Visitor Follows Direction



Service Obtained

8.0 SYSTEM MODULES

8.1 User Management Module

Allows administrators to:

- Create users.
- Disable users.
- Assign roles.
- Reset passwords.
- Manage permissions.
- Possible roles:
 - System Administrator
 - Registry Officer
 - Records Officer
 - Department Officer
 - Supervisor
 - Management
 - Public/Visitor

8.2 Incoming Correspondence Module

Functions:

- Register incoming documents.
- Record sender.
- Record date received.
- Record subject.
- Attach scanned document.

- Assign department.
- Generate reference number.
- Track status.

8.3 Outgoing Correspondence Module

Functions:

- Register outgoing letters.
- Record recipient.
- Record dispatch date.
- Generate reference number.
- Record delivery method.
- Store a digital copy.

8.4 Digital Filing Module

- Provides:
- File creation.
- File classification.
- Document attachment.
- Digital folders.
- File indexing.
- Search and retrieval.

8.5 File Tracking Module

Shows:

- Who has the file?
-
- File number
- Officer
- Department

- Date issued
- Date returned
- Current status

8.6 Document Search Module

Users can search by:

- Reference number
- File number
- Subject
- Sender
- Recipient
- Date
- Department
- Taxpayer identifier where authorized

8.7 Office Directory Module

Contains:

- Office name
- Office number
- Department
- Floor
- Wing
- Location
- Contact information
- Services

8.8 Service Guide Module

For every service, the system can display:

- Service Name
- Office

- Location
- Required documents
- Procedure
- Estimated process steps
- Relevant instructions

8.9 Building Navigation Module

Provides a simplified digital map showing:

- Reception
- Registry
- Offices
- Corridors
- Stairs
- Elevators
- Waiting areas
- Emergency exits

8.10 Public Display Module

The large screen can display:

TANZANIA REVENUE AUTHORITY	
WELCOME TO TRA	
Q	SEARCH FOR SERVICE OR OFFICE
OFFICE DIRECTORY	
REGISTRY SERVICES	
SERVICE PROCEDURES	

	BUILDING MAP	
	ANNOUNCEMENTS	
	1	
	SCAN QR CODE 	

9.0 DATABASE DESIGN

A relational database can be used.

Main tables

1. Users

Field	Description
-------	-------------

user_id	Primary Key
---------	-------------

full_name	User name
-----------	-----------

username	Login username
----------	----------------

password_hash	Encrypted password
---------------	--------------------

role_id	User role
---------	-----------

department_id	Department
---------------	------------

status	Active/Inactive
--------	-----------------

2. Roles

Field	Description
-------	-------------

role_id	Primary Key
---------	-------------

role_name	Role
-----------	------

permissions	Authorized functions
-------------	----------------------

3. Departments

Field Description

department_id Primary Key

department_name Department

description Description

4. Offices

Field Description

office_id Primary Key

office_name Office

office_number Room number

floor Floor

location Location

department_id Department

5. Services

Field Description

service_id Primary Key

service_name Service

description Description

office_id Responsible office

procedure Procedure

requirements Requirements

6. Documents

Field Description

document_id Primary Key

reference_no Reference

subject Subject

sender Sender

date_received Date

document_type Type

file_path Digital document

status Status

7. File_Movement

Field Description

movement_id Primary Key

document_id Document

from_user Sender

to_user Receiver

date_sent Date

date_received Date

status Status

8. Announcements

Field Description

announcement_id Primary Key

title Title

message Message

start_date	Start
end_date	End
status	Status

9. Audit_Log

Field	Description
-------	-------------

log_id	Primary Key
--------	-------------

user_id	User
---------	------

action	Action
--------	--------

record_id	Record
-----------	--------

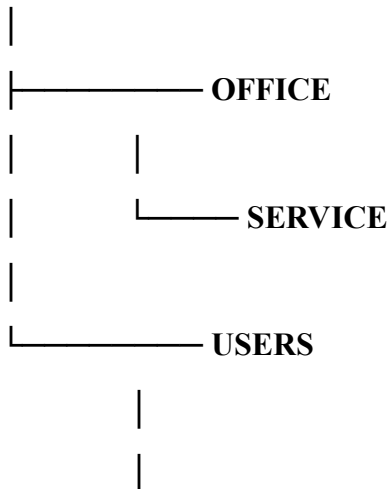
timestamp	Time
-----------	------

IP/device	Device information
-----------	--------------------

10.0 DATABASE RELATIONSHIP

Simplified ERD:

DEPARTMENT



DOCUMENT — FILE_MOVEMENT

```
|
|
└── AUDIT_LOG
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USERS — ROLES

ANNOUNCEMENTS

11.0 SCREEN LAYOUT

The public screen should have a simple interface because visitors may have different levels of computer literacy.

Home Screen

TRA INFORMATION			
SEARCH OFFICE OR SERVICE			
SERVICES	OFFICES	BUILDING MAP	
PROCEDURES	REGISTRY	ANNOUNCEMENTS	

 SCAN QR CODE

Search Result

SEARCH: TIN

SERVICE: TIN SERVICES

OFFICE: Taxpayer Services

OFFICE NO: 105

FLOOR: 1st Floor

LOCATION: Left Wing

PROCEDURE:

1. Go to Office 105
2. Present required documents
3. Follow officer instructions

[VIEW MAP] [SCAN QR]

12.0 ADMINISTRATIVE DASHBOARD

The administrator should have a dashboard showing:

ADMINISTRATOR DASHBOARD			
Documents	Pending	Completed	
245	38	207	
Offices	Services	Users	
52	126	84	

Recent Documents

File Movement

Pending Actions

Announcements

System Alerts

13.0 SECURITY REQUIREMENTS

Because the system will handle official records, security should be a major component.

The system should include:

- Username/password authentication.
- Role-based access control.
- Strong password policies.
- Encryption.
- Secure document storage.
- Audit trails.
- Automatic session timeout.
- Regular database backups.
- Disaster recovery.
- Antivirus/security controls.
- Controlled access to confidential records.
- Logging of important user activities.
- Public users should not have access to internal/confidential registry records.

14.0 IMPLEMENTATION PLAN

Implementation should be done in phases.

Phase 1: Requirements Analysis

Duration: 2–4 weeks

Activities:

- Interview Registry staff.
- Identify current workflows.
- Identify departments and offices.
- Identify services.

- Identify problems.
- Collect system requirements.

Phase 2: System Design

Duration: 3–4 weeks

Activities:

- Database design.
- System architecture.
- UI design.
- Building map design.
- Workflow design.
- Security design.

Phase 3: Development

- Duration: 6–10 weeks
- Development of Registry module.
- Office directory.
- Service guide.
- Search.
- File tracking.
- Administration.
- Public display.
- QR functionality.

Phase 4: Data Preparation

Duration: 3–6 weeks

Activities:

- Enter office information.
- Enter office numbers.
- Enter locations.

- Enter services.
- Enter procedures.
- Prepare digital records.
- Scan priority documents.

Phase 5: Testing

Duration: 2–3 weeks

Testing should include:

- Functional testing.
- Security testing.
- Database testing.
- User acceptance testing.
- Screen/display testing.
- Performance testing.

Phase 6: Staff Training

Duration: 1–2 weeks

Training for:

- Registry staff.
- Administrators.
- Department officers.
- Management.

Phase 7: Pilot Implementation

The system should first be introduced in one selected TRA office/building.

Problems identified during the pilot can then be corrected before wider implementation.

Phase 8: Full Implementation

After successful pilot testing:

- Deploy the system.
- Install public screens.

- Configure network.
- Migrate required records.
- Provide technical support.

15.0 ESTIMATED COST

The actual cost would depend heavily on whether TRA uses existing infrastructure and whether the system is developed internally or procured externally. A pilot implementation could have an indicative budget such as:

Item	Estimated Cost (TZS)
System analysis & design	3,000,000
Software development	12,000,000
Database/server setup	5,000,000
Large digital display	4,000,000
Touchscreen/kiosk	3,500,000
Scanner	1,500,000
Network equipment	2,000,000
Backup/storage	2,500,000
Building map/digital directory	2,000,000
QR implementation	500,000
Staff training	2,000,000
Testing/deployment	2,000,000
Maintenance/contingency	4,000,000
Estimated Total	44,000,000 TZS

Note: These are preliminary planning estimates, not vendor quotations. A formal procurement exercise should determine the actual costs.

16.0 EXPECTED BENEFITS

16.1 Benefits to TRA

- Increased efficiency
- Digital workflows can reduce unnecessary manual work and make documents easier to route and retrieve.
- Better accountability
- The system can record who received, transferred or processed a document.
- Improved document security
- Access controls and audit logs can help protect sensitive information.
- Reduced paper dependence
- More records can be stored digitally where appropriate.
- Faster retrieval
- Authorized staff can search for documents instead of manually searching through shelves.

17.0 BENEFITS TO TAXPAYERS AND VISITORS

Visitors can:

- Find offices quickly.
- Know the correct office number.
- Know which floor to visit.
- See the location on a map.
- Understand procedures.
- Know required documents.
- Reduce unnecessary movement.
- Access information through QR codes.
- This can create a better customer experience.

18.0 BENEFITS TO MANAGEMENT

Management can receive reports such as:

- Number of incoming documents.
- Number of outgoing documents.
- Pending documents.

- Completed documents.
- Average processing time.
- File movement statistics.
- Department workload.
- Registry performance.

*This allows management to make decisions using real operational data rather than relying only on manual reports.

19.0 KEY PERFORMANCE INDICATORS (KPIs)

The project can measure success using:

KPI Target

- Document registration time Reduce significantly
- File retrieval time Reduce significantly
- Lost/misplaced files Minimize
- Visitor direction inquiries Reduce
- Digital documents Increase
- File tracking coverage 100% for registered movements
- System availability $\geq 99\%$ target
- User adoption $\geq 90\%$ of trained staff
- Visitor satisfaction Improve

Exact targets should be established during the requirements and baseline assessment phase.

20.0 RISKS AND MITIGATION

Risk Mitigation

- Resistance to change Staff training and change management
- System failure Backup and disaster recovery
- Cybersecurity threats Security controls and monitoring

- Power interruption UPS and backup power
- Poor internet Reliable internal network
- Incorrect data Validation and data quality controls
- Poor adoption User training and support
- Equipment failure Maintenance and replacement plan
- Unauthorized access Role-based permissions
- Data loss Automated backups

21.0 CONCLUSION

The proposed Digital Registry Management, Office Directory and Visitor Guidance System would provide a structured approach to modernizing Registry operations and improving information access within TRA offices.

Rather than simply replacing paper registers with computers, the project proposes a broader digital transformation of the Registry workflow.

The system would connect:

Digital Registry + Document Tracking + Office Directory + Service Guide + Building Map + Public Screen + QR Code + Management Reports

The public screen would become a digital information point, while the internal system would provide authorized TRA staff with tools for registration, tracking, retrieval and management of records.

A phased implementation beginning with a pilot TRA building is recommended. After evaluating the pilot, the system could be improved and progressively expanded to other suitable TRA offices.

Proposed project name

TRA-DIRECT: Digital Registry, Office Directory and Visitor Guidance System

Core idea:

“From Manual Registry to Smart Digital Registry and Guided Service.”